From: DNV		**Date:**	12-12-2025
To: Sweco		**Prep. By:**	**P.J Schutte (DNV)**

Copied to:
Givan Verveen, Emile Strydom

Subject: **Budgetary Proposal – NEN 3654:2023 Step 3 Detailed Interference Study**

Dear Givan/Emile

Following our review of the preliminary interference studies (Step 1 & 2) prepared by Sweco for the WNL OOST pipeline conversion project, DNV is pleased to present this budgetary proposal.

As discussed, the preliminary Unity Checks (Step 2) have indicated potential exceedances of the NEN 3654:2023 safety limits regarding touch voltages and AC corrosion risk. Consequently, a **Step 3 Detailed Calculation** (compliant with NEN 3654 Annex C) is required to quantify the actual interference levels and determine if mitigation is necessary.

DNV will utilize industry-standard electromagnetic field simulation software (e.g., CDEGS or equivalent) to model the complex interactions between the high-voltage infrastructure and the Gasunie pipelines.

Below is the scope of work and budgetary estimate split per project section.

1 PROJECT 1: TRACÉ ANGERLO – RAVENSTEIN (LEIDING A-505)

Scope of Work:

Construction of a detailed simulation model to calculate induced voltages during steady-state operation (AC corrosion risk) and single-phase fault conditions (Touch voltage safety).

- **High Voltage (HS) Modeling:**
    - **380 kV Overhead Line:** TenneT DTC380-DOD380 (Doetinchem-Dodewaard). *Note: This involves a significant parallel run of approx. 24.7 km requiring detailed tower geometry modeling.*
    - **50 kV Overhead Line:** TenneT BML-NM (Bemmel-Nijmegen).
- **Medium Voltage (MS) Modeling:**
    - Modeling of 4 distinct 10 kV cable parallelisms (Liander B, C, ProRail I, II).
- **Deliverables:** Step 3 Report including voltage profiles and NEN 3654 compliance check.

2 PROJECT 2: TRACÉ RAVENSTEIN – BOXTEL (LEIDING A-526)

Scope of Work:

This section represents the highest complexity due to the density of high-voltage crossings and parallelisms.

- **High Voltage (HS) Modeling:**
  - **380 kV Overhead Line:** TenneT HGL380-ZL380 (Hengelo-Zwolle).
  - **110 kV Overhead Line:** TenneT RT110-NVD110.
  - **380 kV Overhead Line:** TenneT DTC380-HGL380. *Note: Although split into sections A, B, and C in the preliminary study, DNV will model this as a continuous complex interaction.*
  - **150 kV Cable:** TenneT DTC150-ZP150.
  - **380 kV Overhead Line:** TenneT DOD380-DTC380.
- **Medium Voltage (MS) Modeling:**
  - Modeling of 3 distinct 10 kV cable parallelisms (Enexis A, Liander A, B).
- **Deliverables:** Step 3 Report including voltage profiles and NEN 3654 compliance check.

3 PROJECT 3: TRACÉ ZWEEKHORST – OMMEN (LEIDING A-505)

Scope of Work:

Detailed modeling of specific high-voltage cable interactions.

- **High Voltage (HS) Modeling:**
  - **150 kV Connection:** TenneT OS150-UD150.
  - **150 kV Connection:** TenneT OST150-BXT150 (Cables 1 + 2).
- **Medium Voltage (MS) Modeling:**
  - Modeling of 1 distinct 10 kV cable parallelism (Liander B).
- **Deliverables:** Step 3 Report including voltage profiles and NEN 3654 compliance check.

4 SUMMARY OF COSTS

Project Section	Estimated Budget Range
1. Angerlo – Ravenstein	€ 30 000
2. Ravenstein – Boxtel	€ 32 000
3. Zweekhorst – Ommen	€ 27 000
ESTIMATE	**€ 89 000**
Estimate for additional future mitigation (Future Work for phase 2)	**€ 15 000/project**

5 IMPORTANT NOTES & ASSUMPTIONS

1. **Input Data:** This proposal assumes the provision of necessary data from the grid operators (TenneT/Liander/Enexis/ProRail), specifically:
   - Fault current data (Single phase short circuit levels).
   - Load flow data (Normal operation currents).
   - Tower geometries (Mastbeelden) and cable configurations.
   - *DNV can assist in drafting the data request sheets for you to send to the operators.*
2. **Thermal Influence:** In line with the Sweco screening, we have **excluded** thermal calculations from this base budget as these are existing pipelines. However, should the operational temperature profile of the pipeline change significantly due to the conversion to Hydrogen, we recommend a high-level review. This can be added as a provisional item upon request.
3. **Mitigation Design:** This budget covers the *calculation* of interference levels (Step 3). If levels exceed NEN 3654 limits, the design of mitigation measures (e.g., grounding grids, gradient control mats) will be quoted separately as a "Phase 2" scope.

We look forward to supporting Gasunie in ensuring the safety and integrity of this critical infrastructure.

Best regards,

PJ Schutte
Principal Consultant EMC
DNV